

SMART INDIA HACKATHON 2026



TITLE PAGE · DHRISTI

Problem Statement ID – SIH26171

Problem Statement Title – On-device Visual Perception
for Light-weight Browser Agents

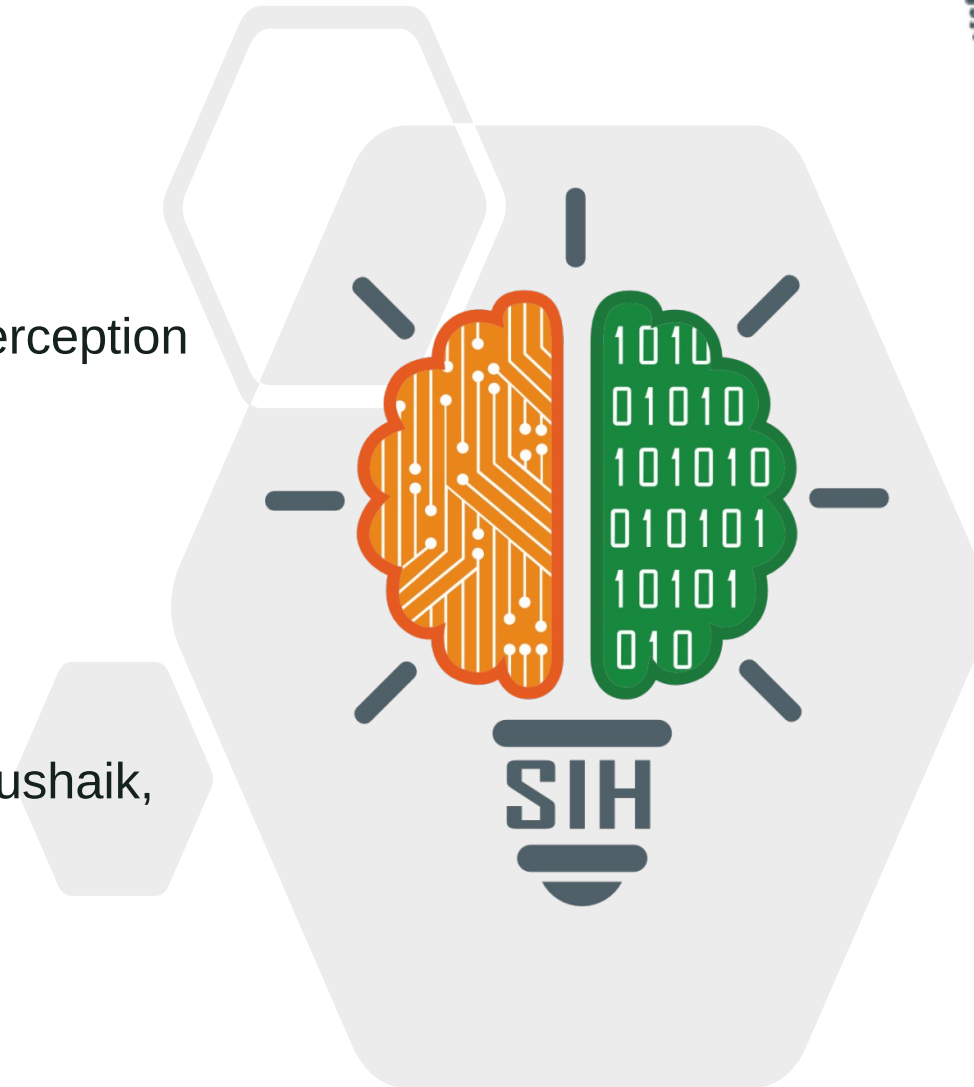
Theme – Smart Automation

PS Category – Software

Team ID – pending registration detail

Team Name (Registered on portal) – pending

RV University · Gopreet, Hiranmayi, Varun, Koushaik,
Francis, Niharika



Proposed Solution (Describe your Idea/Solution/Prototype)

Keep the screen local. Send a protected layout. Review every action.

Detailed explanation of the proposed solution

Local face CV + DOM bounds → opaque regions and approved controls → Qwen2.5 → reviewed action.

How it addresses the problem

A strict pixel-free request excludes source text, images, values and URLs before server reasoning.

Innovation and uniqueness of the solution

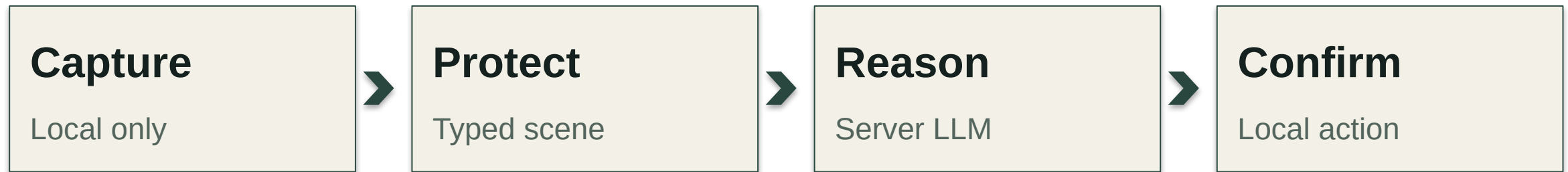
Inspectable egress + expiring action references. Integration difference; no first-ever privacy claim.

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

JavaScript / extension · UltraFace RFB-320 + ONNX WASM · Node.js / Zod
Ollama + Qwen2.5:7B · SQLite audit counts

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Observed: Pending → Review → Request ready for review (synthetic fixture).

Waves 1–4: selective preview, protect loop harness, privacy-only mode; G11 fail honest.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

Local browser + optional Qwen; 119 Node tests pass (measured 2026-09-15).
Privacy-only + Score heuristic; Chromium protect loop harness-verified.

Potential challenges and risks

Real model step: 3,029 ms; full-flow <200 ms target fails (G11).
Face-only CV; WebPIL text retained PII on 58/100; toolbar needs human.

Strategies for overcoming these challenges

Held-out fixtures + privacy-only path + detector cache (Wave4).
Fail closed on sanitize; expiry/cancel recovery evidenced; judge checklist.

Potential impact on the target audience

Help users complete browser tasks with less raw screen exposure.

Benefits of the solution (social, economic, environmental, etc.)

User control

Inspect server-bound context; approve a bounded action.

Deployable locally

Existing laptop + local model; offline operation after assets are installed.

Measure the trade-off

PII coverage vs useful context; task success; latency and client memory.

Expected benefits, not achieved impact. No user study, savings or energy reduction is claimed.

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RESEARCH AND REFERENCES



Details / Links of the reference and research work

Official SIH26171 · problem and 25/20/20/20/15 weights

<https://www.sih.gov.in/sih2026PS>

WebPII / WebRedact · visual PII benchmark (2026 preprint)

<https://arxiv.org/abs/2603.17357>

PrivWeb · local anonymization and user control (2025 preprint)

<https://arxiv.org/abs/2509.11939>

Available but Invisible · typed placeholders (2026 preprint)

<https://arxiv.org/abs/2602.10139>

UltraFace · MIT model; ORT Web · packaged WASM runtime

<https://github.com/Linzaer/Ultra-Light-Fast-Generic-Face-Detector-1MB>

Prototype evidence: Benchmarks/results/prototype-v01-browser.json · model hashes/licenses in Prototype/models/manifest.json

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